Generative and Explainable AI for Crop Disease Detection



By

Muhammad Nabeel Raza
(Registration No: 00000450170)

Department of Computing

School of Electrical Engineering and Computer Science

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

(2025)

Generative and Explainable AI for Crop Disease Detection



By

Muhammad Nabeel Raza

(Registration No: 00000450170)

A thesis submitted to the National University of Sciences and Technology, Islamabad,

in partial fulfillment of the requirements for the degree of

Master of Science in

Data Science

Supervisor: Dr. Rafia Mumtaz

School of Electrical Engineering and Computer Science

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

(2025)

THESIS ACCEPTANCE CERTIFICATE

Certified that final copy of MS Thesis written by Mr Muhammad Nabeel Raza (Registration No. 00000450170), of School of Electrical Engineering and Computer Science has been vetted by undersigned, found complete in all respects as per NUST Statutes/ Regulations/ Masters Policy, is free of plagiarism, errors, and mistakes and is accepted as partial fulfilment for award of Masters degree. It is further certified that necessary amendments as point out by GEC members and foreign/ local evaluators of The scholars have also been incorporated in the said thesis.

Signature:————————————————-

Name of Supervisor: ————————————

Date: ————————————————————

Signature (HoD):————————————————-

Date:————————————————————

Signature (Dean/Principal):————————————

Date: ————————————————————

Form TH-4

National University of Sciences and Technology

MS THESIS WORK

We hereby recommend that the thesis prepared under our supervision by: Mr. **Muhammad Nabeel Raza** Titled: "*Generative and Explainable AI for Crop Disease Detection*" be accepted in partial fulfilled the requirements for the award of MS degree.

Examination Committee Members

1. Name.....Signature: .....

2. Name.....Signature: .....

3. Name.....Signature: .....

4. Name.....Signature: .....

Supervisor's Name.....Signature: .....

Head of Department's Signature.....Date.....

COUNTERSIGNED

Principal/Dean.....Date.....

CERTIFICATE OF APPROVAL

This is to certify that the research work presented in this thesis, entitled "Generative and Explainable AI for Crop Disease Detection" was conducted by Mr. Muhammad Nabeel Raza under the supervision of Dr. Rafia Mumtaz. No part of this thesis has been submitted anywhere else for any other degree. This thesis is submitted to the Department of Computing, School of Electrical Engineering and Computer Science, in partial fulfilment of the requirements for the degree of Master of Science in the field of Data Science, School of Electrical Engineering and Computer Science, National University of Sciences and Technology, Islamabad.

Student Name: ———————————- Signature: ———————————

Examination Committee:

a) External Examiner 1: Name———————- Signature:———————————

(Designation and Office Address)

———————————

b) External Examiner 2: Name———————- Signature:———————————

(Designation and Office Address)

———————————-

Name of Supervisor: ———————————— Signature:———————————

Name of Dean/HOD: ————————————Signature:———————————

AUTHOR'S DECLARATION

I Muhammad Nabeel Raza hereby state that my MS thesis titled "Generative and Explainable AI for Crop Disease Detection" is my own work and has not been submitted previously by me for taking any degree from the National University of Sciences and Technology, Islamabad or anywhere else in the country/ world.

At any time if my statement is found to be incorrect even after I graduate, the university has the right to withdraw my MS degree.

Student Signature:————————————

Name:————————————

Date: ————————————

PLAGIARISM UNDERTAKING

I solemnly declare that the research work presented in the thesis titled "Generative and Explainable AI for Crop Disease Detection" is solely my research work with no significant contribution from any other person. The small contribution/ help wherever taken has been duly acknowledged and that complete thesis has been written by me.

I understand the zero tolerance policy of the HEC and National University of Sciences and Technology (NUST), Islamabad, towards plagiarism. Therefore, I as an author of the above titled thesis declare that no portion of my thesis has been plagiarized and any material used as reference is properly referred/cited.

I undertake that if I am found guilty of any formal plagiarism in the above titled thesis even after award of MS degree, the University reserves the rights to withdraw/revoke my MS degree and that HEC and NUST, Islamabad has the right to publish my name on the HEC/University website on which names of students are placed who submitted plagiarized thesis.

Student Signature:————————————————

Name:————————————————————————

Date: ————————————————————————

This thesis is dedicated to *my beloved parents*

ACKNOWLEDGEMENTS

In the name of ALLAH (SWT) the most Beneficent, the most Merciful, who has given me the strength and ability to complete this thesis. All praises belong to ALLAH (SWT), Lord of the universe. May His blessings be upon the Prophet Muhammad (PBUH) and members of his family and companions.

It is with heartfelt appreciation that I acknowledge my supervisor, Dr. Rafia Mumtaz, along with committee members Dr. Daud Abdullah, Dr. Hashir Kiani, and Ms. Hirra Anwar, for their outstanding guidance, unwavering patience, inspiring motivation, and extensive expertise. Their invaluable support was instrumental in the successful completion of my thesis.

I am deeply appreciative of my parents for their unwavering presence, constantly providing support and encouragement through their invaluable prayers and heartfelt best wishes. Furthermore, I am thankful to Ms Maryam Mohsin for her willingness to promptly annotate the data.

Contents

LIST OF TABLES **V**

LIST OF FIGURES **VII**

LIST OF SYMBOLS, ABBREVIATIONS AND ACRONYMS **VIII**

ABSTRACT **IX**

1 Introduction **1**

1.1 Background 1

1.2 Motivation 2

1.3 Contributions 3

2 Related Work **5**

2.1 Generative AI 5

2.2 Explainable AI 7

2.3 Research Gap 14

3 Materials and Methods **15**

3.1 Data 15

3.2 Exploratory Analysis 16
3.3 Data Processing 17
3.4 Generative AI 18
3.4.1 Generative Adversarial Network 19
3.4.2 Autoregressive Diffusion Models 20
3.4.3 Latent Diffusion Models 21
3.5 Disease Detection 21
3.5.1 ResNet 21
3.5.2 DenseNet 23
3.5.3 MobileNet 24
3.5.4 EfficientNet 25
3.6 Explainable AI 25
3.6.1 Grad-CAM 26
3.6.2 Guided Backpropagation 27
3.6.3 Guided Grad-CAM 28
3.7 Metrics of Evaluation 30
3.7.1 Fréchet Inception Distance 30
3.7.2 Kernel Inception Distance 31
3.7.3 Learned Perceptual Image Patch Similarity 31
3.7.4 Evaluating XAI Techniques 32

4 Results and Discussion **33**

4.1 Synthetic Data Generation 33
4.2 Disease Detection 35
4.3 Explainable AI (XAI) 37

5 Conclusion **42**

BIBLIOGRAPHY **42**

A Appendix A **51**

List of Tables

2.1 Generative AI literature in Plant Leaf Disease Detection 8

2.2 Explainable AI literature in Plant Leaf Disease Detection (Part I) . . . 12

2.3 Explainable AI literature in Plant Leaf Disease Detection (Part II) . . . 13

3.1 Prompts used for text-to-image generation 19

3.2 Model architectures used for training classifiers 22

4.1 Performance comparison of model architectures on extended and non-extended datasets 37

List of Figures

1.1 A leaf sample affected by wheat rust. 2

3.1 Region of Data Acquisition, NARC, Islamabad 16

3.2 Class distribution of the training (a) and testing (b) data 17

3.3 Methodology 18

3.4 Residual Connection to allow better gradient flow in backpropagation . 22

3.5 A dense block in DenseNet where each feature map from previous layers is passed as input to the next layers. 24

3.6 Guided GradCAM from a combination of GradCAM and Guided Backpropagation 29

4.1 Samples from synthetic image generation 34

4.2 Samples from original dataset 35

4.3 FID (a), KID (b), and LPIPS (c) of synthetic data generation step . . . 36

4.4 Train, validation split after synthetic data generation 36

4.5 Confusion Matrix of EfficientNet b2 model trained on extended dataset 38

4.6 Accuracy and F1 score comparison of models on extended and non-extended datasets 39

4.7 Explanation method preference by Class 40

4.8 Example of XAI techniques applied to wheat rust disease detection . . 41

A.1 Confusion Matrix of MobileNet v3 52

A.2 Confusion Matrix of EfficientNet b0 52

A.3 Confusion Matrix of EfficientNet b1 53

A.4 Confusion Matrix of EfficientNet b2 53

A.5 Confusion Matrix of EfficientNet b3 54

A.6 Confusion Matrix of Dense Net 121 54

A.7 Confusion Matrix of ResNet 18 55

A.8 Confusion Matrix of ResNet 34 55

A.9 Confusion Matrix of ResNet 50 56

A.10 Confusion Matrix of ResNet 101 56

LIST OF SYMBOLS, ABBREVIATIONS AND ACRONYMS

XAI — Explainable Artificial Intelligence

IS — Inception Score

FID — Fréchet Inception Distance

KID — Kernel Inception Distance

LPIPS — Learned Perceptual Image Patch Similarity

Grad-CAM — Gradient-weighted Class Activation Mapping

LoRA — Low Rank Adaption

LIME — Local Interpretable Model-agnostic Explanations

ARDM — Autoregressive Diffusion Model

LDM — Latent Diffusion Model

Abstract

In recent literature, the creation of synthetic data and Explainable Artificial Intelligence (XAI) have become prominent, especially in terms of enhancing datasets and explaining model behaviour and biases. Despite this, current research often encounters challenges with the quality of generative data or primarily focuses on explaining models for lesion-based diseases. Diseases like wheat rust often go unattended even though these diseases affect the yield immensely. Explaining model behaviour using traditional methods is also challenging as it requires a fine-grain view of the affected regions. This study aims to fill these gaps by producing high-quality synthetic data via Stable Diffusion and methodically evaluating different XAI methods, namely Grad-CAM, Guided Backpropagation, and Guided Grad-CAM, to shed light on model behaviours. We fine-tuned Stable Diffusion v1.5 using LoRA, resulting in a Fréchet Inception Distance (FID) of 122.21 and a Kernel Inception Distance (KID) of 0.0272 ± 0.0003. Furthermore, in the constructed dataset, the EfficientNet b2 model achieved a classification accuracy of 95.3%. This performance is an improvement of 2.3 percentage points compared to the model trained on the initial dataset, which had an accuracy of 93.01%. In a similar vein, the model's F1 score increased to 0.941, indicating an improvement of 0.026 relative to the F1 score of 0.915 obtained by the model trained on the original dataset. Our analysis reveals that Guided Grad-CAM offers the most efficient

explanations, assisting stakeholders such as farmers in gaining insight into the model's decision-making process, thereby enhancing their trust in the model's outcomes.
Keywords: Stable Diffusion, Explainable AI, Guided Backpropagation, Grad-CAM, Guided Grad-CAM

Chapter 1

Introduction

This chapter presents the issue we aim to address, emphasizing its significance and explaining the distinctiveness and originality of our proposed solution.

1.1 Background

According to the United Nations, the global population has surged significantly, exceeding previous estimates. Since the mid-1900s, it has grown threefold [52, 39]. Nations with lower incomes per capita often experience higher population densities. Despite efforts by governments across the globe to reduce growth rates, the population is anticipated to rise sharply. Consequently, it becomes imperative to secure sufficient food supply to meet the escalating needs.

According to [45], wheat production must be increased by 50% to meet the population's needs by the year 2050. There are numerous factors that contribute to reduced yields, such as the scarcity of cultivable land, insufficient irrigation facilities, climate changes, and outbreaks of diseases. Wheat rust stands out as a predominant factor in

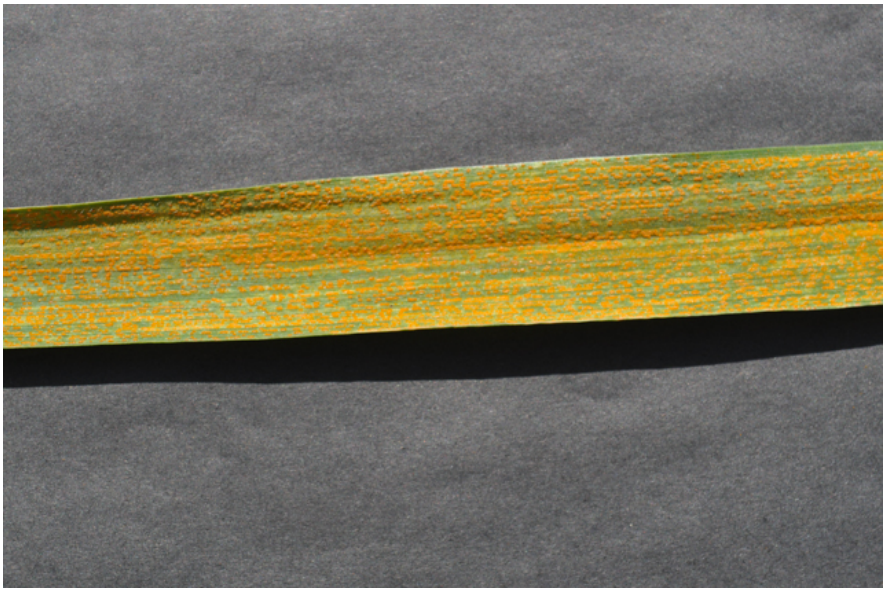


Figure 1.1: A leaf sample affected by wheat rust.

crop loss, leading to yield reductions of up to 20%. Even though wheat is vulnerable to over a hundred different diseases, wheat rust is recognized as the most damaging [40].

The fungus *Puccinia Striiformis* is responsible for inducing stripe rust in wheat. Initially limited to cooler environments, this pathogen has evolved to thrive in warmer areas too. It affects the leaves by blocking sunlight, thereby hindering photosynthesis [45]. Figure 1.1 provides a sample of the disease.

1.2 Motivation

Detecting wheat rust in its initial phases is crucial to mitigate its effects. Conventional methods largely rely on manual labour, which is often inefficient. Numerous studies have explored the application of cutting-edge technologies, such as the Internet of Things (IoT), Unmanned Aerial Vehicles (UAVs), satellite imagery, and Artificial Intelligence (AI), to tackle the challenge of crop diseases.

AI's impact across various fields has been profound, given its wide-ranging adaptability. Despite this, real-world applications remain largely unimplemented. This is frequently attributed to a lack of adequate data or hesitancy in utilizing solutions due to model biases. These issues are often rooted in the unpredictable behaviour inherent to neural networks.

The purpose of this study goes beyond the generation of synthetic data to also explain the results. Based on the research of [44], we incorporate data from the National Agriculture Center (NARC) located in Islamabad, Pakistan. These data categorize wheat rust conditions into three distinct classifications: (I) leaves that are normal and healthy, (II) leaves that exhibit resistance to wheat rust, and (III) leaves that are vulnerable and susceptible to wheat rust.

1.3 Contributions

This research's main contributions can be summarized in the following points:

- **Synthetic Data Generation Framework** – By incorporating Low-Rank Adaptation (LoRA) into a sophisticated stable diffusion model, we produce synthetic datasets that closely resemble the original samples while maintaining low correlation, effectively addressing the common shortcomings of traditional data augmentation techniques.

- **Disease Detection Models** – Multiple designs of Convolutional Neural Networks (CNNs) were developed and rigorously evaluated for their ability to identify diseases. The model demonstrating the highest performance was selected for further interpretability analysis.

- **Comparative Study of Explainability Techniques** – An extensive examination and comparison of various explanation approaches have been conducted, revealing that Grad-CAM, a frequently utilized technique in scholarly studies, falls short in providing detailed interpretations. This highlights the necessity for developing more advanced and all-encompassing methods to enhance interpretability.

This thesis is organized into different chapters reflecting different parts of the research. Chapter 2 highlights the current status of research in both Generative AI and Explainable AI with a target of plat leaf disease detection. Chapter 3 explains the data acquisition, data processing techniques, the models used, and the methodology opted for the process. It also highlights the metrics chosen for evaluation. Chapter 4 presents our findings and different results at different parts of the pipeline. We conclude the thesis in Chapter 5

Chapter 2

Related Work

This chapter emphasizes related studies and identifies areas that have been overlooked. We divide the literature into Two parts, Generative AI and Explainable AI. Summary of the individual literature review is present in Table 2.1 and 2.2 respectively.

2.1 Generative AI

The detection of diseases in plants or crops has been extensively explored, leading to various methodologies and a wide range of publicly accessible datasets. Despite this, there is a notable shortage of data for several crucial crops, and annotating these datasets often incurs significant expense. As a result, researchers frequently resort to data augmentation techniques; however, these can generate images that are too alike, offering only slight improvements [27]. The advent of GAN technology [49, 7, 5] has enabled the generation of synthetic data that closely mirrors the original data while minimizing overly strong correlations.

Initially introduced by [8], Generative Adversarial Networks (GANs) remain a cru-

cial area of investigation with numerous modifications designed to handle a wide array of tasks, including converting text to images, transforming images, filling masked areas in images, or even generating new images without conditions. According to [56], the adaptive weighted GAN (AWGAN) has been employed to effectively manage imbalanced datasets, particularly enhancing classification accuracy to 98.4% as indicated by [9]. Additionally, [10] utilized the adaptive discriminator augmentation of StyleGAN2 (SG2-ADA) to generate images of rice leaves, an approach that minimizes the requirement for extensive training datasets. Despite the success, they observed residual blurriness in the resulting images, necessitating post-processing improvements.

A different study employed WGAN for generating rice leaf images and further refined these images by applying Real-ESRGAN to obtain highly detailed outputs. WGAN is instrumental in reducing the typical training instability found in GANs, whereas Real-ESRGAN is effective in producing high-resolution images [1, 54].

Models based on diffusion have achieved remarkable success in both the quality of images they generate and their training robustness [15, 36]. As reported in [21], synthetic data for citrus plants was produced using Denoising Diffusion Probabilistic Models (DDPM). Similarly, the study by [25] evaluated the RePaint model, a DDPM variant for translating images, against InstaGAN for producing images of tomato and grape leaves, concluding that RePaint achieved a much better FID score of 138.28, whereas InstaGAN scored 206.02.

The study presented in [53] employed a diffusion model rooted in Efficient Channel Attention (ECA) to produce synthetic data through an autoregressive method, intended for the classification of diseases affecting notoginseng panax leaves. This model achieved a Fréchet Inception Distance (FID) score of 42.73. Furthermore, they introduced the Inception-SSNet model designed for feature extraction, which signifi-

cantly boosted classification accuracy from 97.04% to 99.44% when leveraged with the synthetic datasets.

2.2 Explainable AI

GradCAM stands as an example-based XAI technique, widely used in academic studies to detect biases within their models. This method is the preferred choice for elucidating the workings of CNN-based models. In the work by [41], the authors assert that the explanations produced through GradCAM and its variant, Guided GradCAM, exhibit high resolution. In contrast, older approaches like LIME and SHAP are often either computationally taxing or lack stability.

In [23], ResNet, MobileNet, and EfficientNet were fine-tuned for detecting 38 plant diseases, achieving an accuracy of 99.63%. The authors utilized LIME for explainability, demonstrating the specific regions of input images that influenced model decisions. They employed the New Plant Disease Dataset, an augmented version of PlantVillage, and focused on the classification task.

Rakesh et al. [34] adapted Inception v3 and ResNet-9 models to identify 38 different diseases, reaching an accuracy rate of 99.2%. They leveraged LIME for feature interpretation and employed GradCAM to generate heatmaps, relying on the datasets from New Plant Disease and PlantVillage for their analysis.

The study conducted by [22] examined approaches for explaining the process of detecting diseases in tomato foliage. By refining the architectures of EfficientNetB3, Xception, and MobileNetV2 models on the PlantVillage Tomato dataset, they reached an accuracy level of 99.3%. To ensure data balance, augmentation techniques were applied. Furthermore, to enhance the interpretability of the models, Saliency maps

Table 2.1. Generative AI literature in Plant Leaf Disease Detection

Sr. no.	Paper Title	Year	Dataset	Results	Img2Img
1	Enhancing Panax notoginseng Leaf Disease Classification with Inception-SSNet and Image Generation via Improved Diffusion Model	2024	Private (Panax Notoginseng)	FID 42.73 Acc. 97.04% on non-generated images Acc. 99.77% on generated images	false
2	Harnessing the power of diffusion models for plant disease image augmentation	2023	PlantVillage (Tomato + Grape)	FID: RePaint (138.28) and InstaGAN (206.02). KID: RePaint (0.089 ± 0.002) and InstaGAN (0.159 ± 0.004)	true
3	Denoising Diffusion Probabilistic Models and Transfer Learning for citrus disease diagnosis	2023	Private (Citrus)	Acc. 99.8%.	false
4	An Improved Approach to Detection of Rice Leaf Disease with GAN-Based Data Augmentation Pipeline	2023	Mendeley Rice Leaf Disease Dataset (Rice)	FID 26.67 KID of 0.08 mAP of 0.93	false
5	Tomato plant disease detection using transfer learning with C-GAN synthetic images	2021	PlantVillage (Tomato)	IS (2.835 ± 0.228). Acc. 99.51% in 5 classes, Acc. 98.65% in 7 classes, Acc. 97.11% in 10 classes.	false
6	A Hybrid Approach for Plant Disease Detection Using E-GAN and CapsNet	2023	Private + PlantVillage (Grapes)	FID 112.56. Acc. 97.63%.	false
7	Unsupervised image translation using adversarial networks for improved plant disease recognition	2020	Private + Cityscape (Tomato)	FID 47.15 NIMA (4.9 ± 1.89) Acc. 86.1%.	true
8	Grape leaf disease identification with sparse data via generative adversarial networks and convolutional neural networks	2023	PlantVillage (Grapes)	mAP 69.74. Acc. 97.5%.	false
9	A High-Quality Rice Leaf Disease Image Data Augmentation Method Based on a Dual GAN	2023	Private (Rice)	PSNR 24.09 SSIM of 0.63 Acc. 91.65% F1 score. 91.83%.	false
10	A Data Augmented Method for Plant Disease Leaf Image Recognition based on Enhanced GAN Model Network	2023	PlantVillage (Maize)	Acc. 98.4%.	false
11	Attention augmented residual network for tomato disease detection and classification	2021	PlantVillage (Tomato)	SSIM 87.53%. Acc. 98.91%.	false

and GradCAM were utilized, effectively emphasizing areas of interest.

As detailed in [2], the integration of EfficientNetB0 and GradCAM was utilized for the purpose of explainable detection of banana diseases, notably achieving an outstanding accuracy of 99.63% on datasets BLSD and BDT. This research emphasized the generation of heat maps to pinpoint and highlight crucial regions involved in the classification process.

In [31], the authors focused on potato leaf disease detection, classifying four disease types: early-blight, septoria, late-blight, and black-leg. Using a deep ensemble model consisting of CNN, CNN-SVM, and DNN, they achieved an impressive accuracy of 99.98%. The study highlighted the use of LIME and SHAP for explainability, helping improve trust in the model by providing detailed explanations for its predictions. This approach not only outperformed existing models but also offered a faster running time (17.86 s). The dataset used was a private collection from farms in West Bengal, India.

In the study by [28], a novel CNN integrated with K-Nearest Neighbors is presented, specifically designed for classifying plant diseases. This method attains an impressive validation accuracy of up to 99.95% and a perfect AUC of 1 on the PlantVillage dataset. The model's interpretability is enhanced through Grad-CAM and OSA, which provide insights into the classification process. Additionally, t-SNE is utilized to visualize deep neural features, thereby demonstrating the architecture's effectiveness and clarity in performance and interpretative capabilities.

The study by the authors in [33] introduces a hybrid model combining CNN and SVM techniques to effectively classify plant diseases affecting strawberries, peaches, cherries, and soybeans. Their innovative model surpasses the performance of existing pre-trained networks such as VGG16, VGG19, and DenseNet, with an impressive classification accuracy of 99.09%, significantly higher than the 53.82% to 98.8% accu-

racy range observed in the pre-trained networks. To improve model interpretability, Grad-CAM is employed to produce class activation maps, enabling visualization of the specific regions implicated in the disease classification process. Although the results demonstrate high accuracy, the study acknowledges a class imbalance issue within the soybean samples, indicating that accuracy may not be the most suitable metric for assessment. The diverse dataset utilized in this research comprises a blend of private sources, the PlantVillage repository, and the Soybean Diseased Leaf dataset.

In [30], a plant disease classification system using explainable artificial intelligence (XAI) is introduced. This system achieves an impressive accuracy rate of 99.69% when identifying 38 different plant diseases. By utilizing fine-tuned models such as ResNet, EfficientNetB0, and MobileNet in conjunction with the Local Interpretable Model-agnostic Explanations (LIME), the study significantly boosts the interpretability of the system. This approach exemplifies the benefits of explainable AI. The research employed the PlantVillage dataset for its analyses.

In the study by [55], the interpretability of deep learning techniques applied to fruit leaf classification is thoroughly examined. The assessment involves VGG, GoogLeNet, and ResNet models while also introducing a novel ResNet-Attention model enhanced with interpretability tools such as SmoothGrad, LIME, and GradCAM. The ResNet model demonstrates remarkable accuracies of 99.11%, 99.4%, and 99.89% respectively. Among the interpretability methods, GradCAM is noted for delivering outstanding feature visualization, whereas LIME emphasizes the details of leaf appearance, and SmoothGrad focuses on broader visual elements. The datasets utilized in this research include PlantVillage and various other publicly available fruit leaf image datasets.

In their research, [26] present an innovative Parallel Depth-wise Separable CNN (PDS-CNN) model specifically designed for the identification of diseases affecting mul-

berry leaves, such as rust and spot. This model, which employs merely 0.53 million parameters and consists of 8 layers, achieves a high accuracy rate of 96.06%. The authors incorporated a dataset curated by sericulture specialists and applied the SHAP methodology for model interpretability, thereby providing a streamlined and effective tool for the diagnosis of plant diseases.

Ethiraj et al. [6] developed an innovative model named DenseNet-SVM (DNet-SVM: XAI) for predicting plant diseases. This model integrates DenseNet to extract relevant features and employs SVM for the classification task, while LIME provides transparency in the decision-making process. Remarkably, it attained a 97% accuracy rate in distinguishing between healthy and infected sugarcane leaves. This advancement holds significant potential for farmers as it aids in detecting plant infections early and recommending appropriate pesticide treatments. The study utilized a dataset comprising proprietary images of sugarcane leaves.

In [50], researchers utilized the VGG16 model for categorizing corn leaf diseases, specifically identifying them as healthy, affected by blight, grey spot, or common rust. To improve understanding, they applied Layer-wise Relevance Propagation (LRP) which created heat maps for the images, thus offering a clearer interpretation. This approach achieved an accuracy rate of 94.67%, surpassing the success of earlier models. The dataset employed for the corn leaf analysis was sourced from Kaggle.

Table 2.2. Explainable AI literature in Plant Leaf Disease Detection (Part I)

Sr. No.	Paper Title	Year	Model	XAI Technique	Dataset	Results	Texture-Based
1	Plant Leaf Disease Detection using Transfer Learning and Explainable AI	2022	ResNet, MobileNet, and EfficientNet	LIME	New Plant Disease Dataset	Acc. 99.63%	false
2	Explainable AI for Crop disease detection	2022	Inception v3 and ResNet-9	LIME, GradCAM	New Plant Disease Dataset and Plant Village	Acc. 99.2%	false
3	Explainable AI for Tomato Leaf Disease Detection: Insights into Model Interpretability	2023	EfficientNetB3, Xception, and MobileNetV2	GradCAM, Saliency Maps	Plant Village dataset (Tomato)	Acc. 99.3%	false
4	Explainable AI Based framework for Banana Disease Detection	2024	EfficientNetB0	GradCAM	Banana Leaf Spot Diseases (BLSD) and Banana Dataset Tanzania (BDT)	Acc. 99.63%	false
5	A study and comparison of deep learning based potato leaf disease detection and classification techniques using explainable AI	2023	Custom architecture (ensemble)	LIME, SHAP	Private (potatoes)	Acc. 99.98%	false
6	Robust diagnosis and meta visualizations of plant diseases through deep neural architecture with explainable AI	2024	CNN + K-nearest neighbours	GradCAM, OSA	Plant Village	Acc. 99.98%	false

Table 2.3. Explainable AI literature in Plant Leaf Disease Detection (Part II)

Sr. No.	Paper Title	Year	Model	XAI Technique	Dataset	Results	Texture-Based
7	CSXAI: a lightweight 2D CNN-SVM model for detection and classification of various crop diseases with explainable AI visualization	2024	CNN + SVM	GradCAM	Private + Plant Village + Soybean Diseased Leaf Dataset	Acc. 99.09%	false
8	Improving Plant Disease Classification With Deep-Learning-Based Prediction Model Using Explainable Artificial Intelligence	2021	ResNet, EfficientNetB0, and MobileNet	LIME	Plant Village	Acc. 99.69%	false
9	Explainable Deep Learning Study for Leaf Disease Classification	2022	VGG, GoogLeNet, and ResNet	SmoothGrad, LIME, GradCAM	Plant Village and Fruit leaf dataset	Acc. 99.89%	false
10	Explainable deep learning model for automatic mulberry leaf disease classification	2023	Custom architecture Parallel Depth-wise Separable CNN (PDS-CNN)	SHAP	Private (Mulberry leaves)	Acc. 96.06%	false
11	A deep learning-based approach for early detection of disease in sugarcane plants: an explainable artificial intelligence model	2024	DenseNet + SVM	LIME	Private (Sugarcane)	Acc. 97%	false
12	Corn leaf disease: insightful diagnosis using VGG16 empowered by explainable AI	2024	VGG16	LRP	Corn or Maize Dataset (Kaggle)	Acc. 94.67%	true

2.3 Research Gap

Across a wide range of prior research, including the studies cited earlier, there is a predominant focus on generating synthetic data and interpreting classification outcomes. Several areas lacking thorough investigation have been identified:

- Few studies concentrate on classifying diseases that utilize artificially created data centered on texture. This presents a challenge due to the complexity involved in accurately portraying texture within the generated images [12].
- The development of detailed explanations is essential during the process of generating synthetic data to accurately identify potential biases. Without these explanations, there is a risk that models could unintentionally assimilate residual noise from the synthesized images, which could compromise their integrity and validity.
- The majority of studies within the XAI (Explainable Artificial Intelligence) field predominantly focus on diseases characterized by the presence of spots or lesions. These types of conditions are typically more straightforward to elucidate compared to those diseases associated with texture changes, such as wheat rust.

Chapter 3

Materials and Methods

In this chapter, we will provide a comprehensive explanation of the various methods used for data collection, detail the specifics of data analysis, and describe in detail the pre-processing techniques that we have utilized. The overall methodology of this study is summarized in Figure 3.3.

3.1 Data

The dataset of wheat stripe rust was gathered from fields at the National Agriculture Research Center (NARC), in Islamabad, Pakistan. Optical images were captured using a Samsung Galaxy A31's rear camera throughout various stages of wheat development. Rust disease typically surfaces in late February and persists until April. Wheat leaves were positioned on a screen with the camera maintained 4-6 inches above, capturing images with the camera directly above and facing down.

The data acquisition process begins with no background present below the leaves. To improve the focus of the images on the wheat stripe rust disease, a whiteboard or

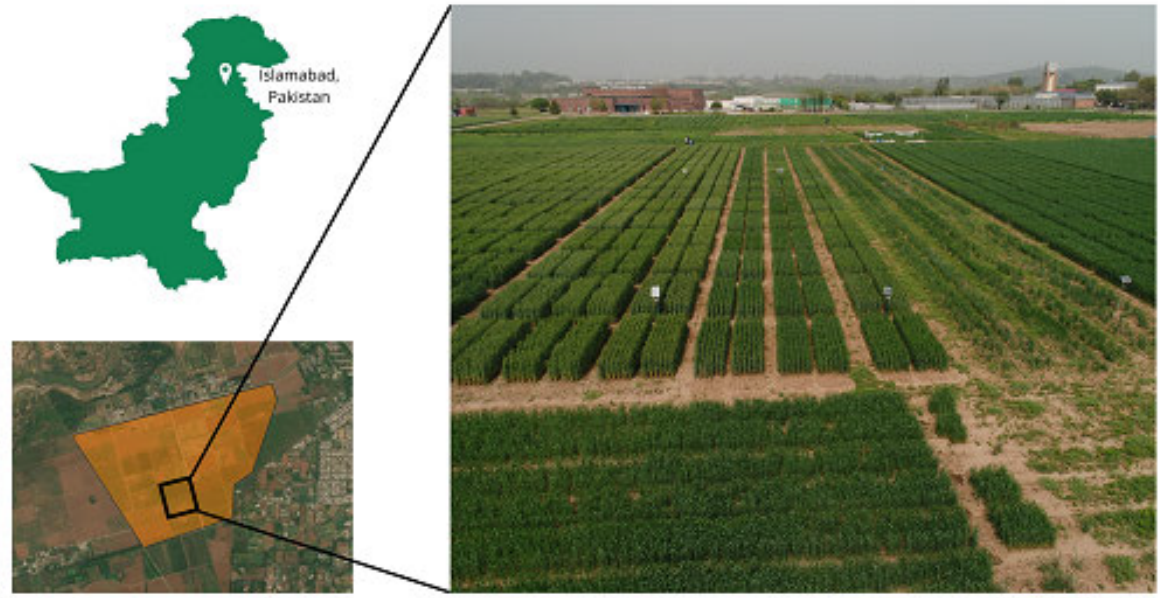


Figure 3.1: Region of Data Acquisition, NARC, Islamabad

blackboard is later introduced, which serves to reduce extraneous background noise effectively. The images are then cropped to a central resolution of 224 x 224 pixels, followed by the application of minimax normalization. This step is crucial, as it strengthens the model's resilience against artifacts while preserving the existing background. Notably, we deliberately choose not to subtract the background, as this allows us to investigate potential biases in the model, especially those associated with the influence of the background environment.

3.2 Exploratory Analysis

We begin the exploratory analysis by looking at the data distribution of the training samples. Figure 3.2 (a) shows the distribution of the training data. As you can see, the class Resistant is less represented in comparison to Susceptible and Healthy. This

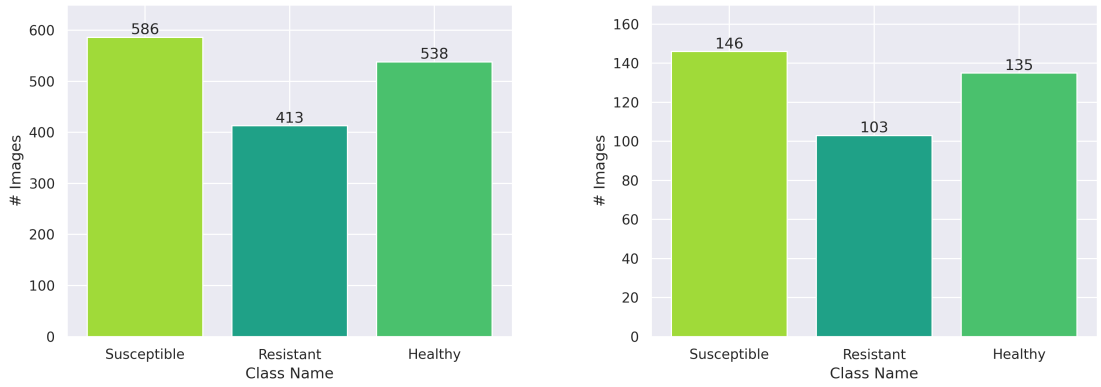


Figure 3.2: Class distribution of the training (a) and testing (b) data

is expected as it is harder to collect resistant samples in nature. Same distribution is reflected in the testing data as well 3.2 (b).

The dimensions of the images, both in terms of size and orientation, exhibited variation due to their capture via a mobile phone. The width of these images is characterized by a median value of 360 pixels, with the range stretching from a maximum of 640 pixels down to a minimum of 319 pixels. In contrast, the height dimension of the images is defined by a median and maximum value of 480 pixels, while the minimum recorded height stands at 425 pixels.

3.3 Data Processing

In order to thoroughly examine biases present in models, we opted not to eliminate the background, as it could itself represent a biasing element. To standardize our dataset, we uniformly cropped the images from the center to a dimension of 256×256 pixels. This step ensured that unwanted artifacts, such as thumbs or fingers, were excluded from the analysis. This methodological choice was introduced after noticing that such artifacts were appearing in generated images. By centering the images, we

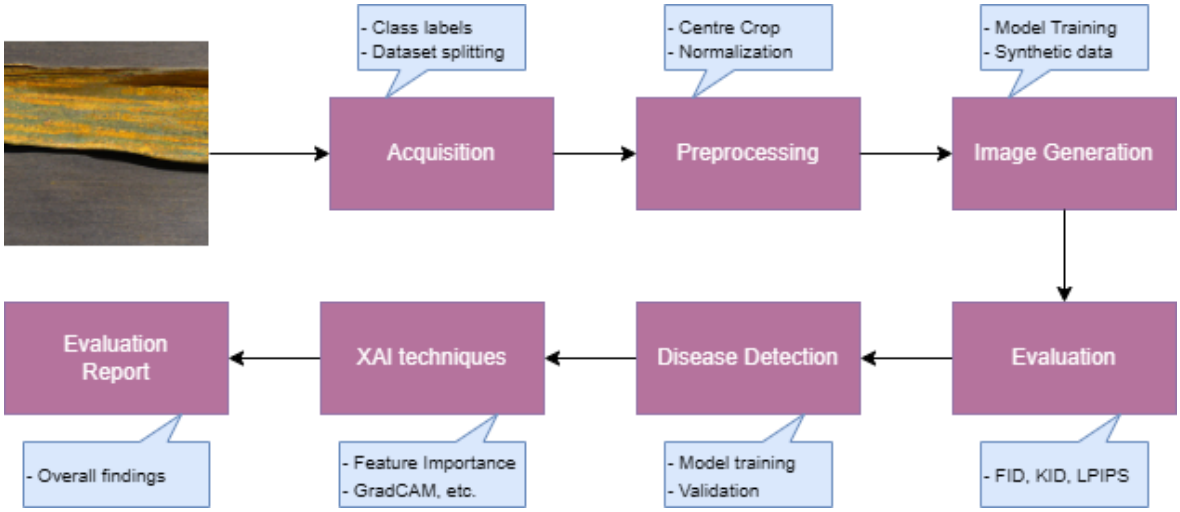


Figure 3.3: Methodology

aimed to facilitate the models in accurately capturing the generative characteristics of the images.

3.4 Generative AI

For synthetic data generation, we use Stable Diffusion v1.5 due to its widespread application and validation in numerous studies, although more advanced models have emerged recently. To begin our method, we apply LoRA (Low-Rank Adaptation) to achieve fine-tuning of parameters with enhanced efficiency, as detailed in [18]. We choose a rank of 4 as it provides an optimal balance between the use of computational resources and the performance of the model. Training LoRA takes less compute and only requires a 3.05 MB or extra storage for one adapter.

We train one LoRA adapter for each class to tackle the problem of context leakage, where a model might generate a different class even if it was specifically prompted to generate one. Please consult Table 3.1 for detailed text prompts. Using different

Class	Text prompt
Healthy	A healthy wheat leaf sample
Resistant	A wheat rust resistant leaf sample
Susceptible	A wheat rust susceptible leaf sample

Table 3.1. Prompts used for text-to-image generation

seed values, randomly selecting inference steps between 25 to 50, we can generate unique images, resulting in samples that exhibit lower correlation than those obtained through conventional linear data augmentation techniques, such as random cropping and rotation. This approach allows us to generate images with variations.

Among the various generative model options, we have selected for Stable Diffusion, which belongs to the category of Latent Diffusion Models. This decision was driven by two primary factors: the superior quality of images it generates and the enhanced control it provides over the image generation process. In subsequent sections, we will discuss the constraints and drawbacks associated with other widely used generative models and explain why they are less ideal for our purposes.

3.4.1 Generative Adversarial Network

Generative Adversarial Networks (GANs) have been a major interest ever since they were introduced by Ian Goodfellow [8]. They work on the principle of adversarial training where a generative model tries to generate images that can fool a discriminator into believing they are the real images. Due to this adversarial training, GANs are notoriously hard to train and are prone to mode collapse.

Numerous studies have been conducted to address this, and WGAN has proposed the creative use of GANs that use the Wasserstein distance as the loss [1]. Unfortunately, these solutions are still not comparable to the later models. Another issue that

GANs face is a lack of control over image generation. The image being generated in vanilla GANs is entirely dependent on the way the initial noisy sample was created and the model itself. Newer studies introduced more control over this by introducing changes to the architecture, such as CycleGAN, CGAN, and AWGAN [58, 24, 9].

3.4.2 Autoregressive Diffusion Models

Diffusion models operate by incrementally introducing noise into an image and training a neural network to reverse this progression. This approach allows the models to approximate the probability distribution underlying the initial data. As a result, they are capable of generating novel samples. The phase where noise is incrementally introduced to the data is known as the diffusion process, whereas the phase responsible for noise reduction, aiming at reconstructing the initial data, is termed as the reverse diffusion process [46].

Autoregressive models pair autoregressive priors with discrete patches, utilizing models like PixelCNN [35] to learn from these patches. In contrast to typical methods, Autoregressive Diffusion Models (ARDMs) incorporate diffusion-like training, optimizing select likelihood terms per step, thereby cutting down generations needed and enhancing sample quality while preserving model richness [14].

ARDMs performed better than GANs and being likelihood models, they did not face the mode collapse issue. The problem with ARDMs is that they require billions of parameters and hundreds of GPU days to train them.

3.4.3 Latent Diffusion Models

While ARDMs solve the latent space problem, the memory requirements and autoregressive nature make them difficult to adopt. Latent Diffusion Models (LDMs) solve this by using their latent space approach. Rather than working in the pixel space, LDMs encode the image in a latent space using a pre-trained autoencoder [36].

LDMs learned to denoise in this latent space, which is much smaller compared to the pixel space, making it less computationally expensive. Additionally, LDMs also incorporated a cross-attention mechanism that gives flexible control over the synthesis via external inputs such as text, bounding boxes, depth maps, etc. [36, 32].

3.5 Disease Detection

For the identification of diseases using classifiers that rely on Convolutional Neural Networks (CNNs), we use the architectural frameworks detailed in Table 3.2. The models undergo a comprehensive fine-tuning process, incorporating both synthetic datasets and real-world data to enhance their learning capabilities. After this refinement phase, the most promising model is selected for applying XAI techniques, ensuring interpretability and reliability. The models chosen for this part are some conventional CNN models well known for their performance in different benchmarks. List of these models in given in Table 3.2. We share a brief introduction to these models in the next sections.

3.5.1 ResNet

Deep convolutional neural networks (CNNs) have demonstrated remarkable success in image recognition tasks, especially with the emergence of AlexNet, with deeper

Sr. no.	Model	Size
1.	MobileNet V3	22.1M
2.	EfficientNet B0	21.4M
3.	EfficientNet B1	31.5M
4.	EfficientNet B2	36.8M
5.	EfficientNet B3	49.3M
6.	DenseNet 121	32.3M
7.	ResNet 18	46.8M
8.	ResNet 34	87.3M
9.	ResNet 50	102M
10.	ResNet 101	179M

Table 3.2. Model architectures used for training classifiers

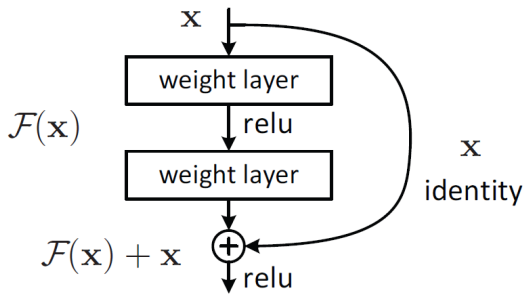


Figure 3.4: Residual Connection to allow better gradient flow in backpropagation

architectures generally leading to improved feature representation and classification performance [20]. However, increasing the depth of CNNs is challenging due to optimization issues such as vanishing gradients and degradation problems, where deeper networks fail to achieve better accuracy than their shallower counterparts. He et al. (2015) addressed these challenges by introducing Residual Networks (ResNets), a deep learning architecture that leverages residual learning to enable training of extremely deep networks [11].

The concept behind ResNet revolves around residual learning, where the network

focuses on learning residual functions in relation to the inputs rather than a straightforward mapping from inputs to outputs. This approach is realized by incorporating identity shortcut connections that cross over one or more layers, facilitating more efficient gradient flow throughout deep networks. These shortcut connections guarantee that the deeper layers can, at a minimum, learn identity mappings, thereby avoiding the decline in accuracy typically observed in conventional deep networks. ResNet's architectural design has made it possible to successfully train networks exceeding depths of 100 layers, achieving cutting-edge performance on renowned benchmark datasets like ImageNet and COCO. Furthermore, the framework for residual learning has become a pivotal methodology in the field of deep learning, impacting a variety of applications beyond just image classification, such as object detection and image segmentation.

3.5.2 DenseNet

Although ResNet introduced residual connections to alleviate vanishing gradients and enable very deep networks, DenseNet takes this concept further by establishing direct connections between all layers within a block. Instead of summing feature maps like ResNet, DenseNet concatenates them, ensuring that each layer retains the information from all its preceding layers. This dense connectivity fosters better reuse of features, stronger gradient flow, and greater efficiency of parameters, leading to better training stability and generalization [19].

Using this connectivity pattern, DenseNet requires significantly fewer parameters than traditional deep networks. Each layer contributes only a small set of new feature maps while reusing previous computations, reducing redundancy. This architecture has demonstrated state-of-the-art performance on benchmark datasets while maintaining computational efficiency. Furthermore, the implicit deep supervision provided by

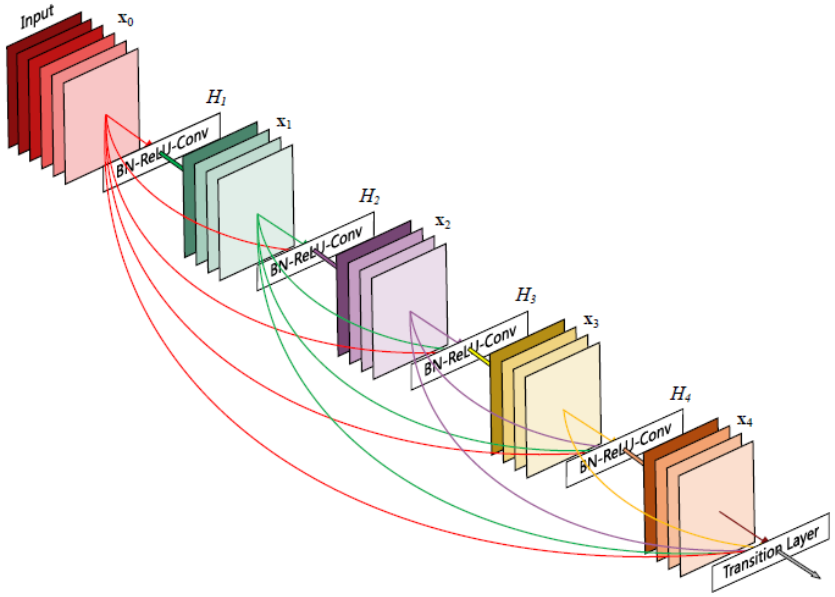


Figure 3.5: A dense block in DenseNet where each feature map from previous layers is passed as input to the next layers.

direct connections to earlier layers improves gradient propagation, making DenseNets particularly well-suited for deep learning applications in image classification and beyond.

3.5.3 MobileNet

While ResNet and DenseNet focused on improving accuracy through deep architectures and efficient gradient flow, MobileNet addresses the challenge of deploying CNNs on mobile and edge devices. Unlike conventional architectures, which prioritize depth and parameter reuse, MobileNet introduces depthwise separable convolutions to drastically reduce computational cost while maintaining competitive accuracy. This factorized convolution technique replaces standard convolutions with a depth-wise convolution followed by a point-wise convolution, significantly decreasing the number of parameters

and floating point operations [17].

Building on the success of MobileNet v1, MobileNet v3 further enhances efficiency through neural architecture search (NAS) and novel architectural improvements. It incorporates squeeze-and-excitation (SE) modules, nonlinear activation functions such as hard-swish, and optimized network scaling strategies. These enhancements enable MobileNet v3 to outperform previous versions in both speed and accuracy, making it a preferred choice for applications in real-time image recognition and mobile AI [16]. By balancing efficiency and performance, MobileNets extend the applicability of CNNs to resource-constrained devices, complementing deeper architectures like ResNet and DenseNet in environments where computational power is limited.

3.5.4 EfficientNet

Traditionally, models are arbitrarily scaled by increasing depth or width, often leading to diminishing returns. However, EfficientNet employs a compound scaling method that uniformly balances depth, width, and resolution based on a fixed set of scaling coefficients. This strategy ensures optimal resource utilization, achieving superior accuracy with fewer parameters compared to previous architectures [48].

3.6 Explainable AI

In this study, methodologies including GradCAM, Guided Backpropagation, and Guided GradCAM were used to generate visual explanations for classified images. GradCAM (Gradient-weighted Class Activation Mapping) was utilized to produce class-discriminative localization maps, highlighting relevant features in the input space. Guided backpropagation was applied to obtain detailed gradient visualizations, which

enhance interpretability by focusing on pixel-level contributions. Furthermore, Guided GradCAM, a hybrid approach that combines the aforementioned techniques, provided comprehensive insights by merging coarse localization with fine-grained gradients, facilitating an in-depth understanding of the decision-making process of the model.

3.6.1 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is a method for generating visual explanations for deep convolutional neural network (CNN) predictions by highlighting the important regions in an image that contribute to a model's decision. Unlike earlier techniques such as Class Activation Mapping (CAM), which required modifications to a model's architecture, Grad-CAM works by computing the gradients of the target class (e.g., a specific object category in image classification) with respect to the feature maps of the last convolutional layer. These gradients are then averaged across spatial dimensions to obtain importance weights for each feature map, which are linearly combined to produce a class-specific localization heatmap. By applying a ReLU activation to this heatmap, Grad-CAM ensures that only the characteristics that positively influence the prediction of the target class are highlighted, resulting in an interpretable class-discriminative visualization of the model focus [42].

One of the key advantages of Grad-CAM is its applicability to a wide range of CNN architectures, including those used for classification, image captioning, and visual question answering (VQA), without requiring retraining or modifications. Since it relies on the gradients of the last convolutional layer, the generated heat maps are often coarse and may lack precise localization of objects, especially in complex images with multiple overlapping features. Additionally, Grad-CAM assumes that important spatial information is retained in the last convolutional layer, which may not always

be true for very deep networks where critical information is spread across multiple layers. Another drawback is that it only highlights regions that positively influence the model's decision, which means that it does not explicitly show the features that could lead to incorrect predictions or conflicting evidence. These limitations suggest that, while Grad-CAM improves model interpretability, it should be used alongside other techniques to gain a more complete understanding of the decision-making process of a model.

3.6.2 Guided Backpropagation

Guided Backpropagation is a visualization technique used to understand which parts of an input image contribute most to the prediction of a deep convolutional neural network. Unlike methods that rely on activation mappings, Guided Backpropagation modifies the standard backpropagation process by allowing only positive gradients to pass through the ReLU activation functions while backpropagating errors. This selective filtering of gradients ensures that only features that positively influence a neuron's activation contribute to the visualization. The result is a fine-grained high-resolution visualization that highlights edges and important structures in the input image, making it particularly useful for understanding how lower-layer filters detect fundamental patterns such as edges and textures [47].

Despite its effectiveness in providing high-detail visual explanations, Guided Backpropagation has some notable limitations. Since it does not provide class-specific information, it is not inherently suited for distinguishing between different object classes within an image. Additionally, the method is highly dependent on the architecture of the network, particularly the choice of activation functions and the layers used for visualization. Another limitation is that it does not account for negative influences

on a model's decision, which means that it only visualizes characteristics that contribute positively to activation, potentially missing important inhibitory signals. Due to these limitations, guided backpropagation is often used in conjunction with other interpretability techniques, such as Grad-CAM, to obtain a more comprehensive understanding of the decision-making process of a model.

3.6.3 Guided Grad-CAM

Guided Grad-CAM is a visualization technique that combines the advantages of Grad-CAM and Guided Backpropagation to produce high resolution and class-discriminative explanations for convolutional neural networks (CNN). Grad-CAM generates coarse heat maps by computing the gradients of a target class with respect to feature maps in the last convolutional layer, highlighting the most important spatial regions influencing the model's decision. However, Grad-CAM lacks fine details. Guided Backpropagation, on the other hand, provides high-resolution edge visualizations but is not class discriminative, as it highlights all strong gradients in the image regardless of the class. Guided Grad-CAM combines these methods by elementally multiplying the Grad-CAM heat map with the Guided Backpropagation visualization, producing a refined saliency map that captures both class-specific localization and fine-grained details [43].

Despite its advantages, Guided Grad-CAM has certain limitations. One key drawback is that it relies on both Grad-CAM and Guided Backpropagation, meaning it inherits the limitations of both techniques. While the approach improves interpretability by providing detailed explanations, it may still produce misleading results in cases where the class localization from Grad-CAM is not precise. Additionally, since Guided Grad-CAM is based on backpropagating gradients, it remains sensitive to network architecture choices, such as activation functions and layer depth. Moreover, because

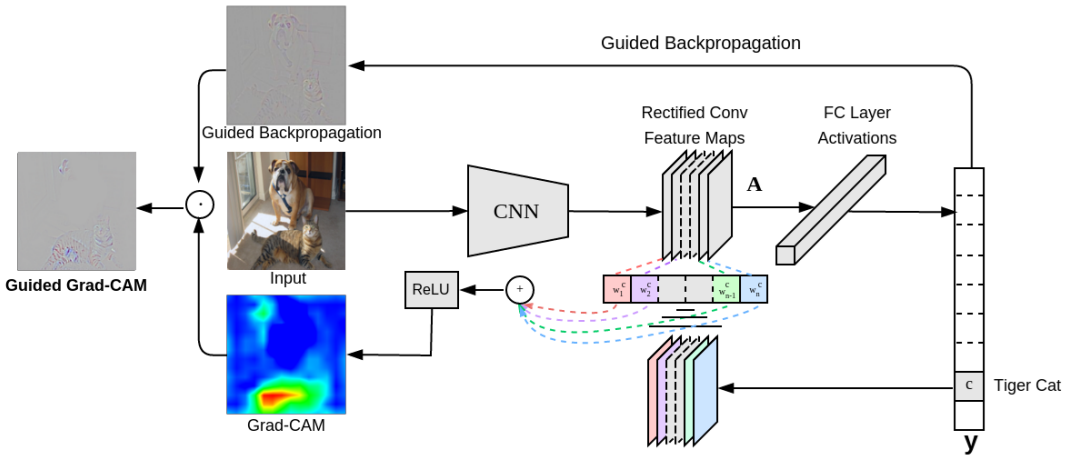


Figure 3.6: Guided GradCAM from a combination of GradCAM and Guided Backpropagation

it only highlights positively contributing features, it does not reveal counter-evidence that might negatively influence a model's decision. Therefore, while Guided Grad-CAM offers a more informative visualization than its components, it should be used alongside other interpretability methods for a more comprehensive understanding of CNN behavior.

3.7 Metrics of Evaluation

This section explores the various metrics used to assess our work. Although evaluating disease detection is straightforward, the generative AI and XAI components require careful decision making. Each generative models is different and utilized for a variety of applications, and since likelihood is used to measure how similar their synthesis is to the real-world samples, it becomes difficult to compare them [51].

3.7.1 Fréchet Inception Distance

Fréchet Inception Distance (FID) is an improved version of the Inception Score. The Inception score uses the KL divergence on features extracted by the Inception models to assess the realism of the generated samples [38]. The equation for Inception Score is given in 3.1. Notice how it does not include comparison with the real-world samples, which could be misleading, especially in case of mode collapse.

$$IS = exp(\mathbb{E}_x KL(p(y|x)||p(y))) \tag{3.1}$$

FID on the other hand compares both real-world samples and the generated ones. FID formula is given in equation 3.2 where $\mathcal{N}(\mu, \sigma)$ is the Gaussian distribution calculated from the real-world samples and $\mathcal{N}(\mu_w, \sigma_w)$ is the Gaussian distribution calculated from the generated samples and the distance between the two distributions is calculated via the Fréchet distance [13].

$$FID = \|\mu - \mu_w\|^2 + tr(\sigma + \sigma_w - 2(\sigma\sigma_w)^{\frac{1}{2}}) \tag{3.2}$$

FID is more consistent with noise levels compared to the Inception score and aligns well with human judgment.

3.7.2 Kernel Inception Distance

FID has a drawback that it relies on Gaussian assumption, which is not reliable with a smaller number of samples. The kernel inception distance (KID) uses kernel methods to capture the data distribution more accurately, making it more robust and less affected by a smaller number of samples [4].

$$\begin{aligned} KID &= MMD(f_{real}, f_{fake})^2 \\ MMD^2 &= \mathbb{E}[k(X, X')] + \mathbb{E}[k(Y, Y')] - 2\mathbb{E}[k(X, Y)] \\ k(x, y) &= (\gamma * x^T y + coef)^{degree} \end{aligned} \tag{3.3}$$

The formula for KID is given in equation 3.3 where MMD is the maximum mean discrepancy and f_{real}, f_{fake} are the Inception features for real and fake data. MMD calculates the distance between the mean embeddings of the two distributions where k is the polynomial kernel function. This kernel function projects the features in a higher dimension where their similarities can be analyzed.

3.7.3 Learned Perceptual Image Patch Similarity

Learned Perceptual Image Patch Similarity (LPIPS) measures the similarity between two image patches with features extracted by a pretrained model, which was VGG in the original study [57]. Generally, FID and KID are the preferred metrics, but here we use LPIPS to measure the similarity between the generated and real-world sample data.

3.7.4 Evaluating XAI Techniques

Assessing methods for Explainable AI (XAI) poses significant challenges. Unlike Generative AI metrics, we lack the assistance of pre-trained models in this evaluation process. As highlighted by David et al. in [29], the Co-12 framework introduces twelve desirable attributes for effective explanations, such as Completeness, Correctness, and Compactness, among others. To accurately assess these techniques, specific transformations are necessary. For example, in evaluating completeness, one typically removes certain data features to observe and analyze the resulting explanation. Thus, these evaluation methods become difficult to compute for high-dimensional data such as images.

Therefore, we compared our findings with the help of human evaluators. We compare the map generated by each technique in Section 3.6. The evaluators judge the explanation based on its high quality of it. We define high-quality generations as those that cover the most rust-affected regions of the leaf and do not highlight unwanted parts, such as the background and healthy regions. One person made the annotations and another person verified them.

Chapter 4

Results and Discussion

This section provides a detailed examination of the results and findings obtained using the methodology described in Section 3. We divide this section into three parts, starting with Synthetic Data Generation 4.1 to explain our findings on training and the generation of fake data for training. Then we have the disease detection section 4.2 specifically for evaluating detection models. And finally, we present our human preference results of XAI techniques in section 4.3.

4.1 Synthetic Data Generation

Our implementation's initial phase involves fine-tuning LoRA adapters for each category of wheat rust disease with Stable Diffusion v1.5. Figure 4.1 presents some instances of the generated images. These generated images look comparable to the original samples in Figure 4.2. To our knowledge, no one in the literature focused on generating realistic images that captured the texture and morphology of the leaves.

We achieved a Fréchet Inception Distance (FID) of 122.2. As mentioned in 3.7.1,

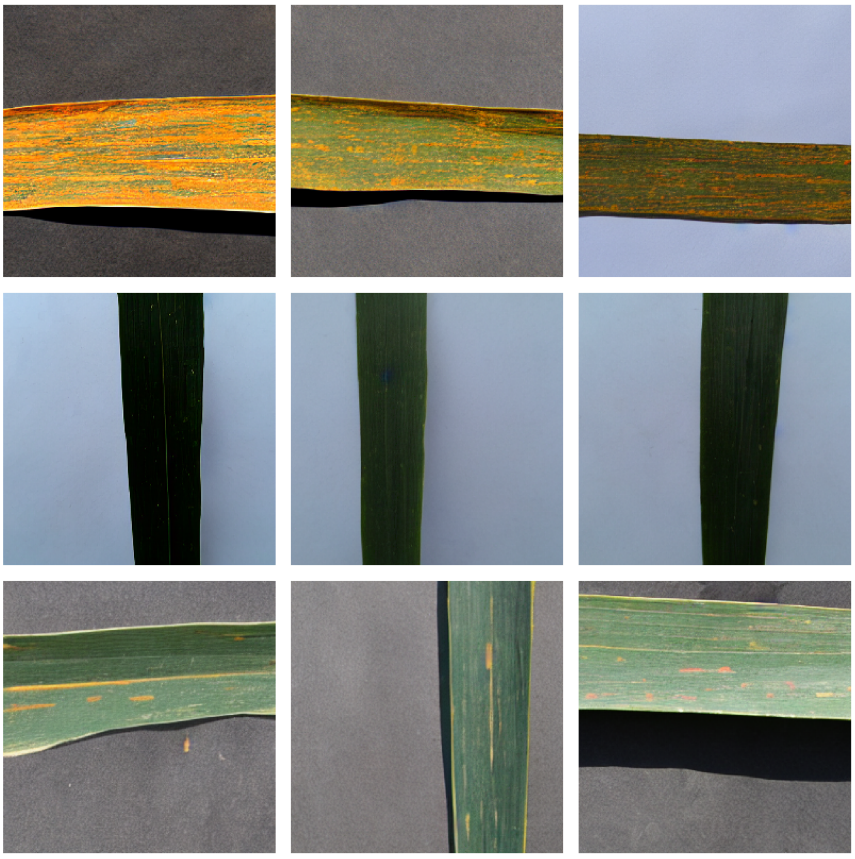


Figure 4.1: Samples from synthetic image generation

FID is a subjective measure and despite its widespread use in research papers for performance metrics, such as in [53, 25, 10] and many more, it is not entirely suitable for small datasets due to its parametric nature. Therefore, for a more comprehensive evaluation, we also assessed our generated samples using the kernel inception distance (KID). We report a KID of 0.0272 ± 0.0003, which is state of the art. To observe diversity, we compute the learned perceptual image patch similarity (LPIPS). Our results show an LPIPS of 0.3298 indicating that neither the images generated were too similar to the original ones nor they were too different.

As can be seen in Figure 3.2, the susceptible and healthy classes are overrepresented

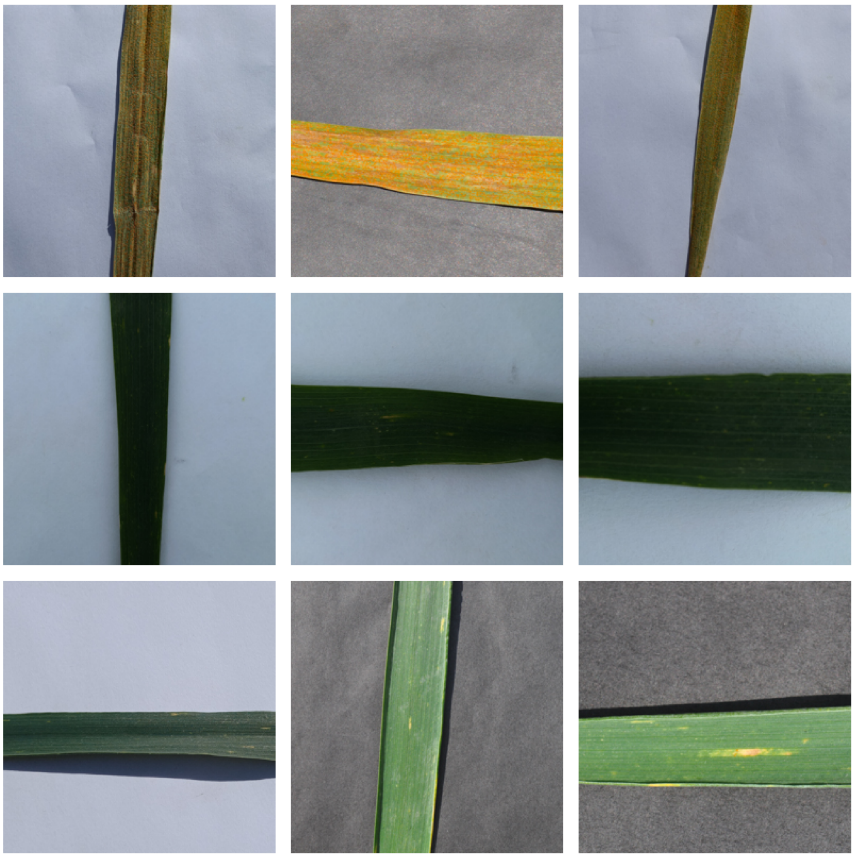


Figure 4.2: Samples from original dataset

compared to the resistant class. After synthesizing data with fine-tuned Stable Diffusion, we round up each class distribution to be equal (Figure 4.4). The test set was omitted from this generation process, as our goal is to ensure that the final evaluation is based on an assessment that relies on authentic, real-world data.

4.2 Disease Detection

Training of the detection models was done with both an extended data set and a non-extended data set for comparison. All models were trained under similar conditions. We keep a small batch size of 16 to have more variations and avoid overfitting and

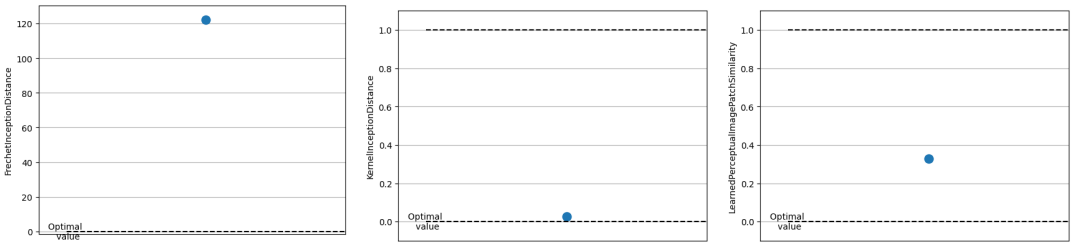


Figure 4.3: FID (a), KID (b), and LPIPS (c) of synthetic data generation step

maximum epochs of 30 with early stopping enabled with a relaxation of 5 epochs. The initial learning rate was set at 0.01, which we reduced by a factor of 10 each time the validation loss plateaued. More details are available in the attached code[1] which will be made public at the time of publishing. The performance evaluation of each model is mentioned in Table 4.1.

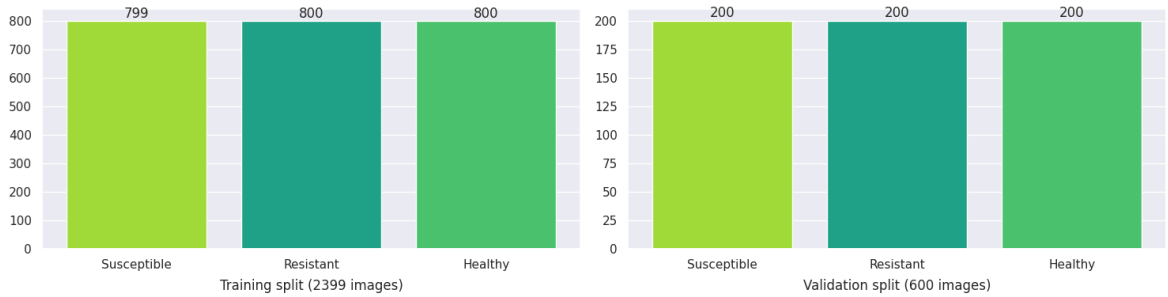


Figure 4.4: Train, validation split after synthetic data generation

The evaluation revealed that among various models, EfficientNet b2 achieved the highest performance when trained on the extended dataset, prompting its selection for the generation of more detailed explanations. Analysis of the confusion matrix shown in Figure 4.5 indicates that most classification errors occurred between the Resistant and Susceptible categories. For the confusion matrix of rest of the models, please refer to Appendix A.

Comparison with other models makes it clear that not all models experience the

[1] `https://github.com/nabeelraza-7/wheat-rust-detection-with-XAI`

Sr. No.	Model	Extended		Non-extended	
		Accuracy	**F1 Score**	**Accuracy**	**F1 Score**
1	MobileNet V3	94.112	0.925	92.164	0.9112
2	EfficientNet B0	93.313	0.9221	93.436	0.9224
3	EfficientNet B1	93.491	0.9179	**93.412**	0.9175
4	EfficientNet B2	**95.292**	**0.9405**	93.014	0.9146
5	EfficientNet B3	93.691	0.9179	94.231	**0.9362**
6	DenseNet 121	93.175	0.9221	93.014	0.9185
7	ResNet 18	95.176	0.9385	94.357	0.9308
8	ResNet 34	93.803	0.9229	93.723	0.9247
9	ResNet 50	93.102	0.9174	93.419	0.9238
10	ResNet 101	93.088	0.919	94.112	0.9275

Table 4.1. Performance comparison of model architectures on extended and non-extended datasets

same type of increase in performance from generated data. This indicates that these models are sensitive to the noise present in the images generated from the extended data. Refer to Figure 4.6 for visuals.

4.3 Explainable AI (XAI)

Despite the availability of various ways to assess XAI methodologies developed by scientists, most of the publications reviewed in this analysis overlooked them. The primary reasons are the difficulties posed by high-dimensional data, such as images, and the unpredictable results of some XAI techniques generated in each run [29].

Given the limited amount of testing data available, we solicit human feedback on three XAI techniques, asking evaluators to indicate which method provides what they

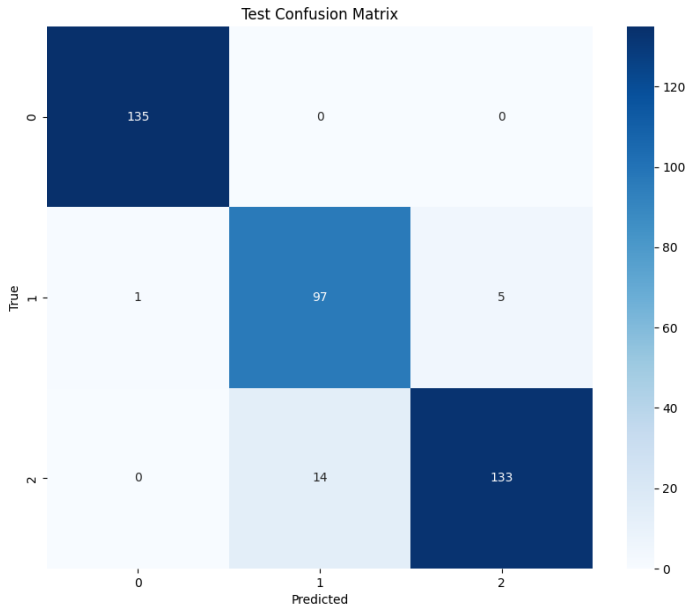


Figure 4.5: Confusion Matrix of EfficientNet b2 model trained on extended dataset

deem to be the highest-quality explanation. Precisely, we characterize a high-quality explanation as one that effectively emphasizes crucial features without extending into irrelevant details. One individual carried out the annotations, which were then assessed by a second individual. Any discrepancies were addressed collaboratively to achieve consensus.

Figure 4.7 illustrates the method preferences categorized by classes. In particular, Guided GradCAM is favoured when the class is healthy and resistant, while GradCAM is chosen for the susceptible category. The heat-map nature of GradCAM covers more area and it is easier for a non-technical person to judge the results. On the other hand, Guided Backpropagation (GBP) tends to generate granular results which are harder to look at. Guided GradCAM, incorporates the best of both worlds giving a granular map with better visual representation than GBP. Some samples are attached in Figure

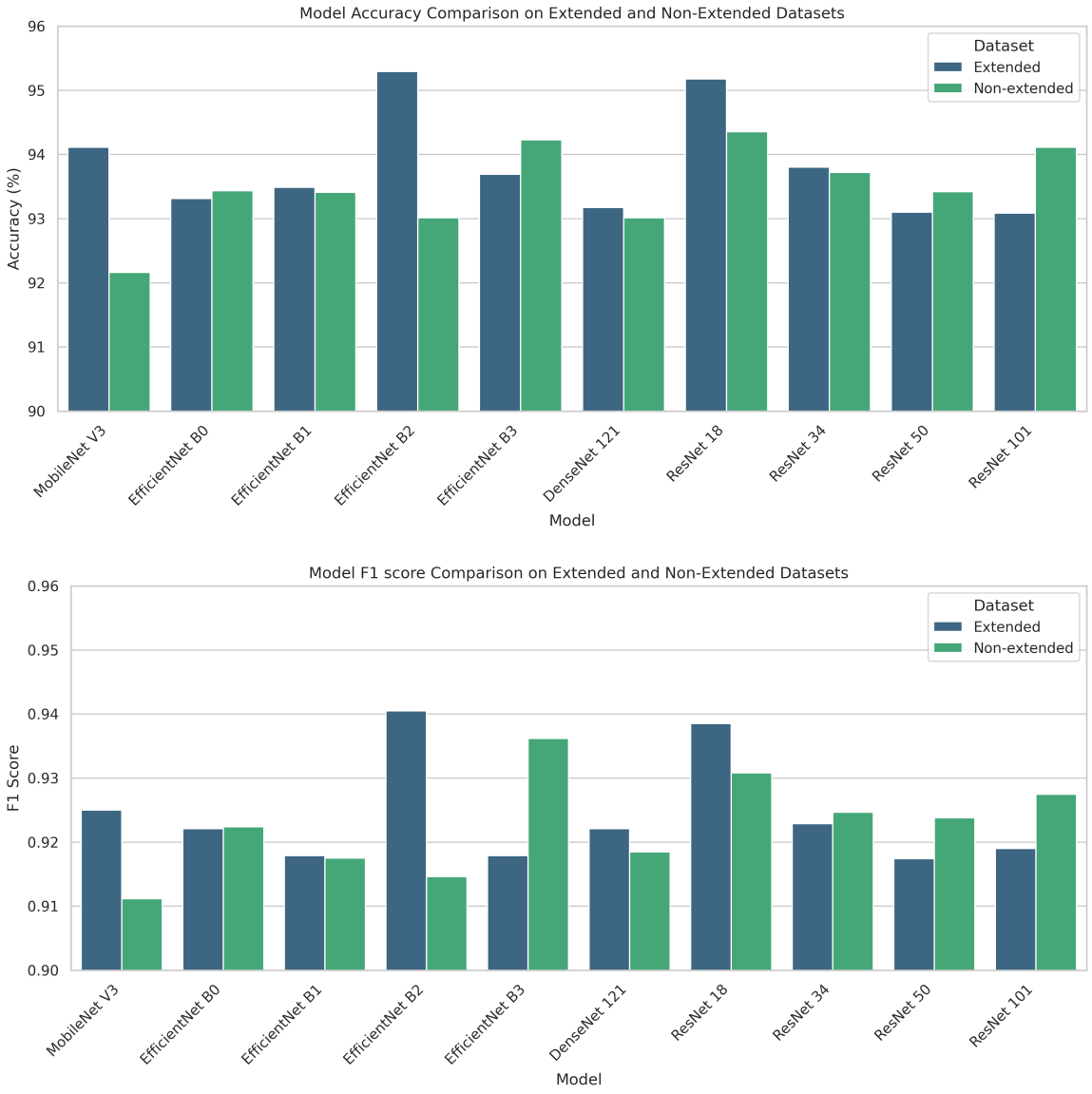


Figure 4.6: Accuracy and F1 score comparison of models on extended and non-extended datasets

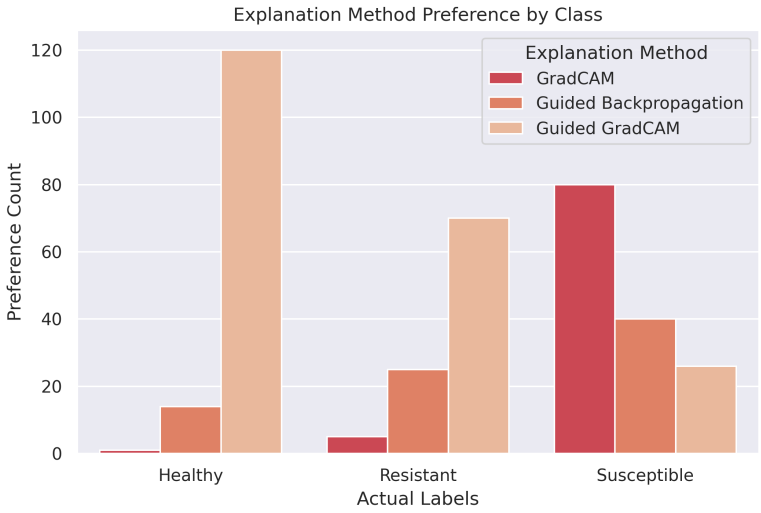


Figure 4.7: Explanation method preference by Class

4.8. We deployed a HuggingFace space to help visualize this[2].

[2] https://huggingface.co/spaces/nabeelraza/wheat-disease-detection-XAI

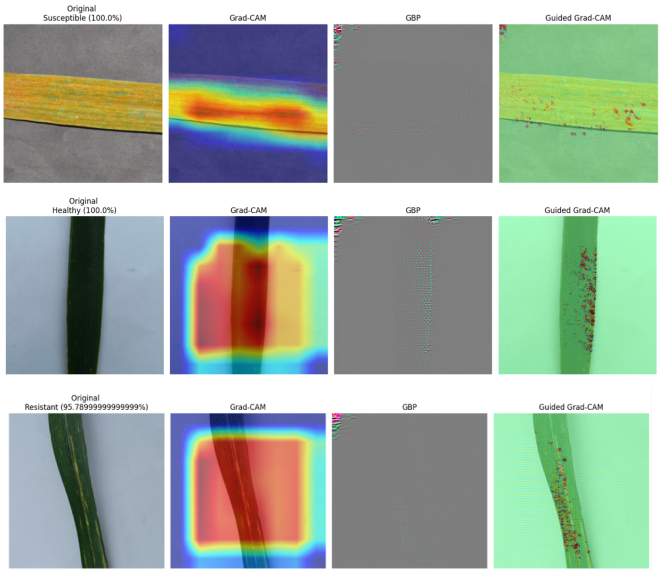


Figure 4.8: Example of XAI techniques applied to wheat rust disease detection

Chapter 5

Conclusion

The preceding results demonstrate that it is possible to augment data utilizing generative AI and address ethical issues regarding AI reliability through the application of XAI. Furthermore, our findings indicate that Guided GradCAM is more appropriate than Guided Backpropagation and GradCAM for diseases characterized by texture, which may fall short in these situations. Despite the challenge of visualizing Guided Backpropagation, there is a risk that GradCAM could mislead by suggesting that the model relies on incorrect features. Future research could focus on improving the visual representation of Guided Backpropagation.

Bibliography

[1] Arjovsky, M., Chintala, S., and Bottou, L. Wasserstein gan, 2017.

[2] Ashoka, S., Pramodha, M., Nyange, R., Anusha, A., Shilpa, G., Chola, C., et al. Explainable ai based framework for banana disease detection. *Research Square* (2024).

[3] Bengamra, S., Zagrouba, E., and Bigand, A. Explainable ai for deep learning based potato leaf disease detection. In *2023 IEEE International Conference on Fuzzy Systems (FUZZ)* (2023), IEEE, pp. 1–6.

[4] Bi'nkowski, M., Sutherland, D. J., Arbel, M., and Gretton, A. Demystifying mmd gans. *arXiv preprint arXiv:1801.01401* (2018).

[5] Eilertsen, G., Tsirikoglou, A., Lundstr"om, C., and Unger, J. Ensembles of gans for synthetic training data generation. *arXiv preprint arXiv:2104.11797* (2021).

[6] Ethiraj, R. P., and Paranjothi, K. A deep learning-based approach for early detection of disease in sugarcane plants: an explainable artificial intelligence model. *Int J Artif Intell ISSN 2252*, 8938 (2024), 8938.

[7] Figueira, A., and Vaz, B. Survey on synthetic data generation, evaluation methods and gans. *Mathematics 10*, 15 (2022), 2733.

[8] Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. Generative adversarial networks, 2014.

[9] Guan, S., Zhao, X., Xue, Y., and Pan, H. Awgan: An adaptive weighting gan approach for oversampling imbalanced datasets. *Information Sciences 663* (2024), 120311.

[10] Haruna, Y., Qin, S., and Mbyamm Kiki, M. J. An improved approach to detection of rice leaf disease with gan-based data augmentation pipeline. *Applied Sciences 13*, 3 (2023), 1346.

[11] He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition, 2015.

[12] Hedjazi, M. A., and Genc, Y. Efficient texture-aware multi-gan for image inpainting. *Knowledge-Based Systems 217* (2021), 106789.

[13] Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., and Hochreiter, S. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems 30* (2017).

[14] Hoogeboom, E., Gritsenko, A. A., Bastings, J., Poole, B., Berg, R. v. d., and Salimans, T. Autoregressive diffusion models. *arXiv preprint arXiv:2110.02037* (2021).

[15] Hoogeboom, E., Gritsenko, A. A., Bastings, J., Poole, B., van den Berg, R., and Salimans, T. Autoregressive diffusion models, 2022.

[16] Howard, A., Sandler, M., Chu, G., Chen, L.-C., Chen, B., Tan, M., Wang, W., Zhu, Y., Pang, R., Vasudevan, V., Le, Q. V., and Adam, H. Searching for mobilenetv3, 2019.

[17] Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., and Adam, H. Mobilenets: Efficient convolutional neural networks for mobile vision applications, 2017.

[18] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. Lora: Low-rank adaptation of large language models, 2021.

[19] Huang, G., Liu, Z., van der Maaten, L., and Weinberger, K. Q. Densely connected convolutional networks, 2018.

[20] Krizhevsky, A., Sutskever, I., and Hinton, G. E. Imagenet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems* (2012), F. Pereira, C. Burges, L. Bottou, and K. Weinberger, Eds., vol. 25, Curran Associates, Inc.

[21] Li, Y., Guo, J., Qiu, H., Chen, F., and Zhang, J. Denoising diffusion probabilistic models and transfer learning for citrus disease diagnosis. *Frontiers in plant science 14* (2023), 1267810.

[22] Mahmud, T., Barua, K., Barua, A., Basnin, N., Das, S., Hossain, M. S., and Andersson, K. Explainable ai for tomato leaf disease detection: Insights into model interpretability. In *2023 26th International Conference on Computer and Information Technology (ICCIT)* (2023), IEEE, pp. 1–6.

[23] Mehedi, M. H. K., Hosain, A. S., Ahmed, S., Promita, S. T., Muna, R. K., Hasan, M., and Reza, M. T. Plant leaf disease detection using transfer learning and explainable ai. In *2022 IEEE 13th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)* (2022), IEEE, pp. 0166–0170.

[24] Mirza, M., and Osindero, S. Conditional generative adversarial nets. *CoRR abs/1411.1784* (2014).

[25] Muhammad, A., Salman, Z., Lee, K., and Han, D. Harnessing the power of diffusion models for plant disease image augmentation. *Frontiers in Plant Science 14* (2023), 1280496.

[26] Nahiduzzaman, M., Chowdhury, M. E., Salam, A., Nahid, E., Ahmed, F., Al-Emadi, N., Ayari, M. A., Khandakar, A., and Haider, J. Explainable deep learning model for automatic mulberry leaf disease classification. *Frontiers in Plant Science 14* (2023), 1175515.

[27] Nanni, L., Paci, M., Brahnam, S., and Lumini, A. Comparison of different image data augmentation approaches. *Journal of imaging 7*, 12 (2021), 254.

[28] Natarajan, S., Chakrabarti, P., and Margala, M. Robust diagnosis and meta visualizations of plant diseases through deep neural architecture with explainable ai. *Scientific Reports 14*, 1 (2024), 13695.

[29] Nauta, M., Trienes, J., Pathak, S., Nguyen, E., Peters, M., Schmitt, Y., Schl"otterer, J., Van Keulen, M., and Seifert, C. From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainable ai. *ACM Computing Surveys 55*, 13s (2023), 1–42.

[30] Nigar, N., Faisal, H. M., Umer, M., Oki, O., and Lukose, J. Improving plant disease classification with deep learning based prediction model using explainable artificial intelligence. *IEEE Access* (2024).

[31] Paul, H., Ghatak, S., Chakraborty, S., Pandey, S. K., Dey, L., Show, D., and Maity, S. A study and comparison of deep learning based potato leaf disease detection and classification techniques using explainable ai. *Multimedia Tools and Applications 83*, 14 (2024), 42485–42518.

[32] Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., and Rombach, R. Sdxl: Improving latent diffusion models for high-resolution image synthesis, 2023.

[33] Prince, R. H., Mamun, A. A., Peyal, H. I., Miraz, S., Nahiduzzaman, M., Khandakar, A., and Ayari, M. A. Csxai: a lightweight 2d cnn-svm model for detection and classification of various crop diseases with explainable ai visualization. *Frontiers in Plant Science 15* (2024), 1412988.

[34] Rakesh, S., and Indiramma, M. Explainable ai for crop disease detection. In *2022 4th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)* (2022), IEEE, pp. 1601–1608.

[35] Razavi, A., Van den Oord, A., and Vinyals, O. Generating diverse high-fidelity images with vq-vae-2. *Advances in neural information processing systems 32* (2019).

[36] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., and Ommer, B. High-resolution image synthesis with latent diffusion models, 2022.

[37] Sagar, S., Javed, M., and Doermann, D. S. Leaf-based plant disease detection and explainable ai. *arXiv preprint arXiv:2404.16833* (2023).

[38] Salimans, T., Goodfellow, I., Zaremba, W., Cheung, V., Radford, A., and Chen, X. Improved techniques for training gans. *Advances in neural information processing systems 29* (2016).

[39] Sara Hertog, Patrick Gerland and John Wilmoth, UN DESA Population Division. Un desa policy brief no. 153: India overtakes china as the world's most populous country, 2023. Accessed: 2025-01-11.

[40] Savary, S., Willocquet, L., Pethybridge, S. J., Esker, P., McRoberts, N., and Nelson, A. The global burden of pathogens and pests on major food crops. *Nat. Ecol. Evol. 3*, 3 (Mar. 2019), 430–439.

[41] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on computer vision* (2017), pp. 618–626.

[42] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. Grad-cam: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision 128*, 2 (Oct. 2019), 336–359.

[43] Selvaraju, R. R., Das, A., Vedantam, R., Cogswell, M., Parikh, D., and Batra, D. Grad-cam: Why did you say that?, 2017.

[44] Shafi, U., Mumtaz, R., Shafaq, Z., Zaidi, S. M. H., Kaifi, M. O., Mahmood, Z., and Zaidi, S. A. R. Wheat rust disease detection techniques: a technical perspective. *J. Plant Dis. Prot. (2006) 129*, 3 (June 2022), 489–504.

[45] Singh, J., Chhabra, B., Raza, A., Yang, S. H., and Sandhu, K. S. Important wheat diseases in the US and their management in the 21st century. *Front. Plant Sci. 13* (2022), 1010191.

[46] Sohl-Dickstein, J., Weiss, E. A., Maheswaranathan, N., and Ganguli, S. Deep unsupervised learning using nonequilibrium thermodynamics, 2015.

[47] Springenberg, J. T., Dosovitskiy, A., Brox, T., and Riedmiller, M. Striving for simplicity: The all convolutional net, 2015.

[48] Tan, M., and Le, Q. V. Efficientnet: Rethinking model scaling for convolutional neural networks, 2020.

[49] Tanaka, F. H. K. D. S., and Aranha, C. Data augmentation using gans. *arXiv preprint arXiv:1904.09135* (2019).

[50] Tariq, M., Ali, U., Abbas, S., Hassan, S., Naqvi, R. A., Khan, M. A., and Jeong, D. Corn leaf disease: insightful diagnosis using vgg16 empowered by explainable ai. *Frontiers in Plant Science 15* (2024), 1402835.

[51] Theis, L., Oord, A. v. d., and Bethge, M. A note on the evaluation of generative models. *arXiv preprint arXiv:1511.01844* (2015).

[52] United Nations. Population, 2025. Accessed: 2025-01-11.

[53] Wang, R., Zhang, X., Yang, Q., Lei, L., Liang, J., and Yang, L. Enhancing panax notoginseng leaf disease classification with inception-ssnet and image generation via improved diffusion model. *Agronomy 14*, 9 (2024), 1982.

[54] Wang, X., Xie, L., Dong, C., and Shan, Y. Real-esrgan: Training real-world blind super-resolution with pure synthetic data, 2021.

[55] Wei, K., Chen, B., Zhang, J., Fan, S., Wu, K., Liu, G., and Chen, D. Explainable deep learning study for leaf disease classification. *Agronomy 12*, 5 (2022), 1035.

[56] Xin, M., Ang, L. W., and Palaniappan, S. A data augmented method for plant disease leaf image recognition based on enhanced gan model network. *Journal of Informatics and Web Engineering 2*, 1 (2023), 1–12.

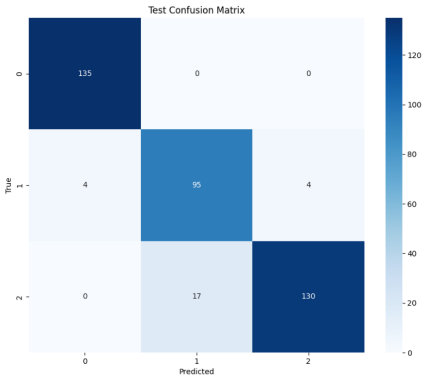
[57] Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (2018), pp. 586–595.

[58] Zhu, J.-Y., Park, T., Isola, P., and Efros, A. A. Unpaired image-to-image translation using cycle-consistent adversarial networks, 2020.

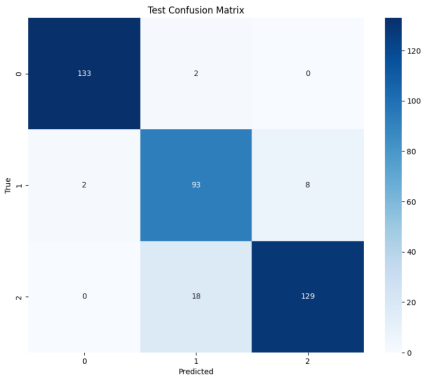
Appendix A

Supplemantary Material

This section contains the confusion matrix for each model that was trained in disease detection.

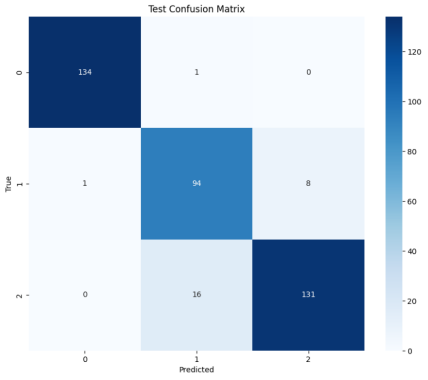


MobileNet v3 | Extended dataset

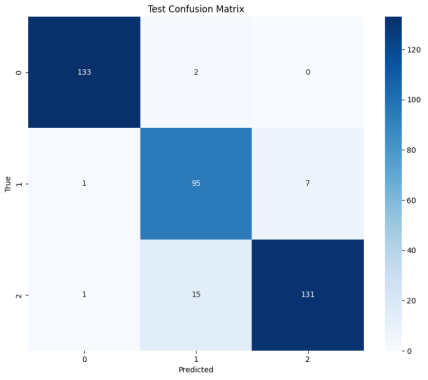


MobileNet v3 | Non-extended dataset

Figure A.1: Confusion Matrix of MobileNet v3

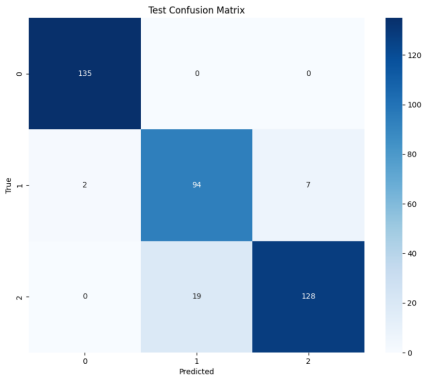


EfficientNet B0 | Extended dataset

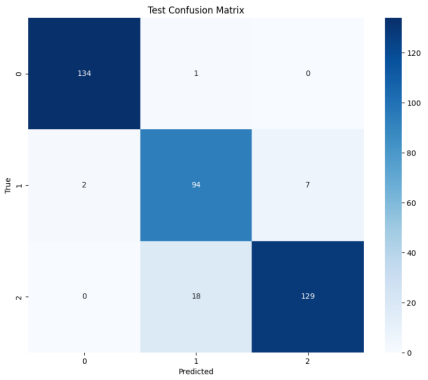


EfficientNet B0 | Non-extended dataset

Figure A.2: Confusion Matrix of EfficientNet b0

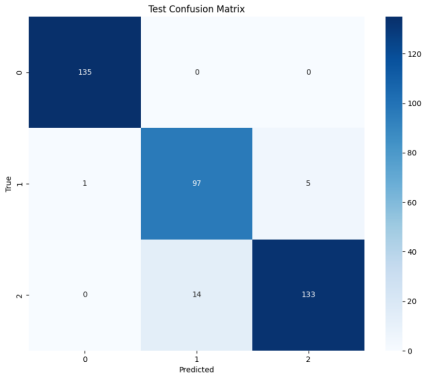


EfficientNet B1 | Extended dataset

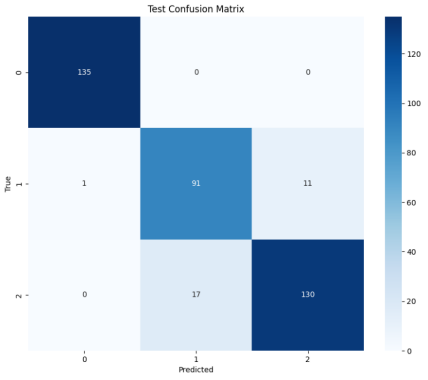


EfficientNet B1 | Non-extended dataset

Figure A.3: Confusion Matrix of EfficientNet b1

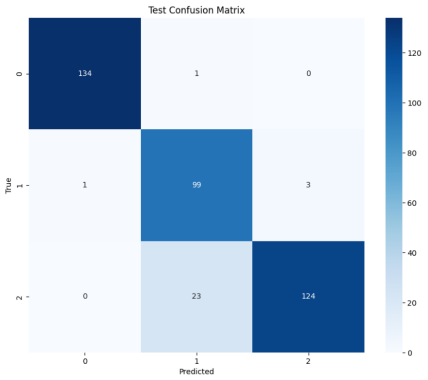


EfficientNet B2 | Extended dataset



EfficientNet B2 | Non-extended dataset

Figure A.4: Confusion Matrix of EfficientNet b2

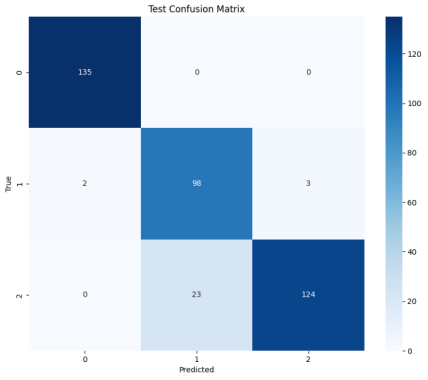


EfficientNet B3 | Extended dataset

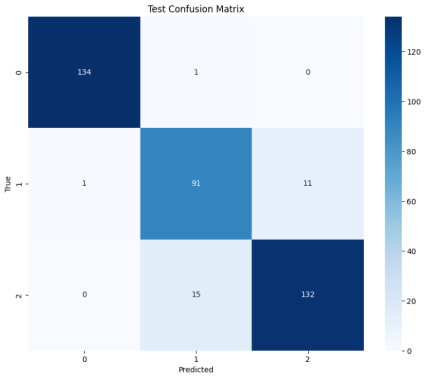


EfficientNet B3 | Non-extended dataset

Figure A.5: Confusion Matrix of EfficientNet b3

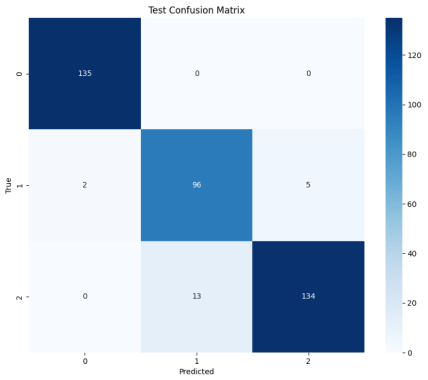


DenseNet | Extended dataset

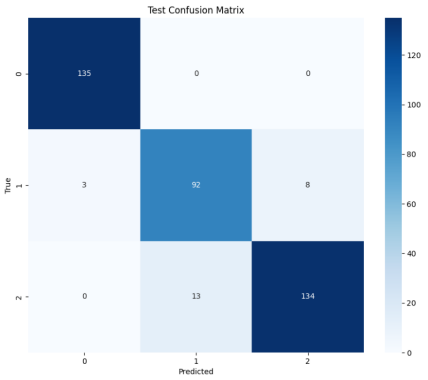


DenseNet | Non-extended dataset

Figure A.6: Confusion Matrix of Dense Net 121

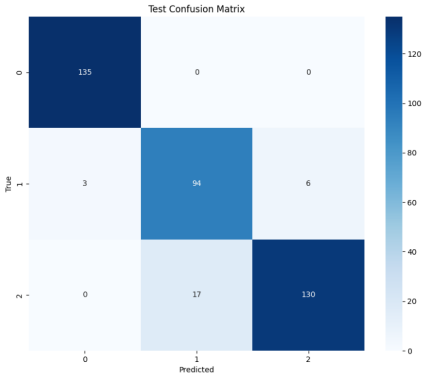


ResNet-18 | Extended dataset

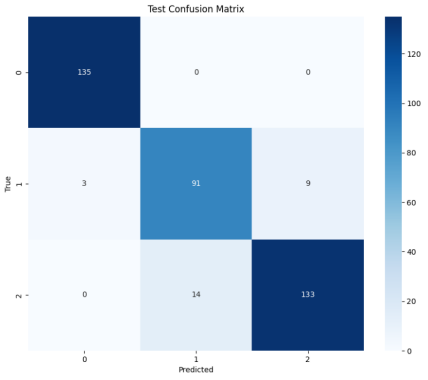


ResNet-18 | Non-extended dataset

Figure A.7: Confusion Matrix of ResNet 18

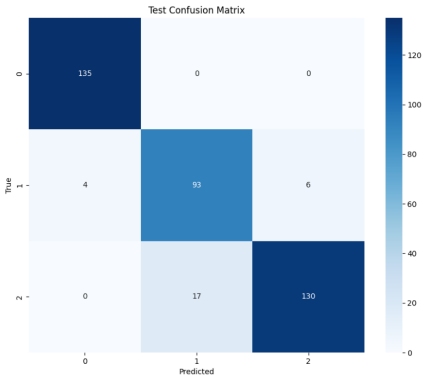


ResNet-34 | Extended dataset

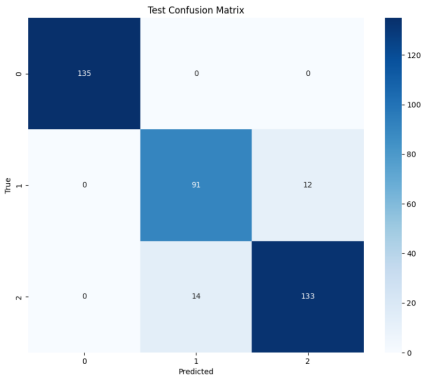


ResNet-34 | Non-extended dataset

Figure A.8: Confusion Matrix of ResNet 34

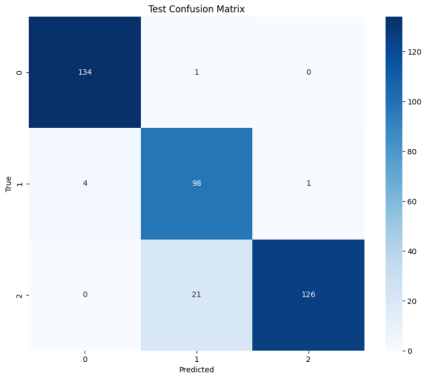


ResNet-50 | Extended dataset

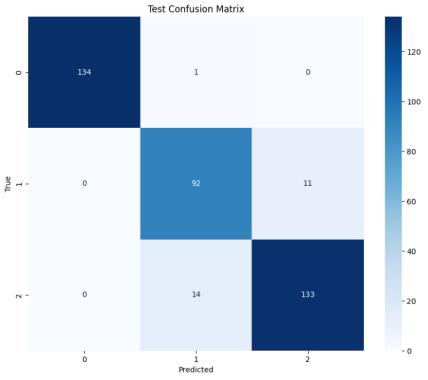


ResNet-50 | Non-extended dataset

Figure A.9: Confusion Matrix of ResNet 50



ResNet-101 | Extended dataset



ResNet-101 | Non-extended dataset

Figure A.10: Confusion Matrix of ResNet 101